

Localization Audio Pipeline

This project demonstrates a structured and scalable localization audio pipeline, simulating real-world workflows used in game production.

Pipeline Overview

1. Dialogue creation and organization (Excel database)
2. Voice recording (Reaper)
3. Audio cleanup (iZotope RX)
4. File naming and organization (Python automation)
5. Integration into Wwise
6. Implementation in Unreal Engine

Project Scope

Game type: Life Simulation (Sims-style)

Characters:

- Male Adult (MA)
- Female Adult (FA)
- System / Narrator (SYS)

Languages:

- English (EN)
- Spanish (ES)

Dialogue Categories

- Greeting (social interaction)
- Complaint (problems, issues)
- Work (tasks, job-related lines)
- Idle (casual thoughts, filler lines)
- Reaction (emotions: surprise, anger, happiness)
- Objective (actions, goals)
- System (UI, instructions)

Naming Convention

A structured and scalable naming convention is used, following common game audio standards.

Format:

VO_<Language><Character><Category><Q/R><ID>_<Variation>.wav

Where:

VO = Voice Over

Language = EN / ES
Character = MA, FA, SYS
Category = Dialogue category
Q = Question, R = Response
ID = 3-digit identifier (001, 002, 003...)
Variation = A, B, C

Examples:

VO_EN_MA_Greeting_Q_001_A.wav
VO_ES_MA_Greeting_Q_001_A.wav

VO_EN_FA_Work_R_012_A.wav
VO_ES_FA_Work_R_012_A.wav

VO_EN_SYS_System_005.wav
VO_ES_SYS_System_005.wav

Important Rules

- The same ID represents the same line across all languages
- File names must remain consistent and error-free
- Use leading zeros for IDs (001, not 1)
- No spaces, only underscores

Recording Guidelines

- Record in 48kHz / 24-bit WAV
- Maintain consistent microphone distance and position
- Avoid clipping and background noise
- Record multiple takes per line (2–3 recommended)
- Keep performance consistent between EN and ES
- Do not apply heavy processing during recording

Audio Processing Guidelines

- Apply de-noise to remove background noise
- Remove clicks, pops, and unwanted artifacts
- Apply basic EQ to ensure clarity
- Normalize loudness across all files
- Ensure consistency across variations and languages

Folder Structure

```
audio_pipeline/  
├─ 01_RAW/  
├─ 02_CLEAN/  
├─ 03_FINAL/  
├─ csv/  
├─ scripts/  
└─ docs/
```

RAW: Original recordings, untouched

CLEAN: Processed audio after RX cleanup

FINAL: Ready for integration into Wwise/Unreal

Final Outcome

This project results in a fully structured and scalable localization audio pipeline, demonstrating the ability to manage dialogue assets from raw recording to in-engine implementation.

Skills Demonstrated

- Audio editing and cleanup (RX workflow)
- Dialogue organization and asset management
- File naming standardization and consistency
- Workflow automation using Python
- Integration-ready audio preparation (Wwise / Unreal)
- Pipeline thinking and production-oriented problem solving